University of Exeter

MTH1001B

Mathematical Structures

On Campus Exam	Closed Book
Additional Materials: None	**Permitted Materials:** Approved Non-Programmable Scientific Calculators Permitted (Green)

Guidelines:

-The **duration** of this paper is 2 Hours
-Word Count: No Word Count
-Write all answers in the answer booklet provided.
-Do not open the exam paper until instructed by the Invigilator.
-Use only black pen or pencil.
-Mobile and electronic devices are not allowed.
-Do not communicate with other candidates during the exam.

Exam Instructions:

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline.

SECTION A

1. (a) Consider the vectors

$$\overrightarrow{v_1} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \quad \overrightarrow{v_2} = \begin{pmatrix} 4 \\ 2 \\ 2 \end{pmatrix}, \quad \overrightarrow{v_3} = \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix}, \quad \overrightarrow{v_4} = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}$$

and

$$\overrightarrow{b} = \begin{pmatrix} 4 \\ 3 \\ 7 \end{pmatrix}.$$

(i) Find a maximal linearly independent sublist of the list of vectors $\overrightarrow{v_1}$, $\overrightarrow{v_2}$, $\overrightarrow{v_3}$, $\overrightarrow{v_4}$.

(ii) Determine whether $\overrightarrow{b}$ can be written as a linear combination of $\overrightarrow{v_1}$, $\overrightarrow{v_2}$, $\overrightarrow{v_3}$, $\overrightarrow{v_4}$ and, if it can, write it in this way. (13)

(b) Let V be a vector space over $\mathbb{R}$. Show that if the vectors $\overrightarrow{w_1}, \overrightarrow{w_2}, \ldots, \overrightarrow{w_n}$ span V then the vectors $2\overrightarrow{w_1}, 4\overrightarrow{w_2}, \ldots, (2^n)\overrightarrow{w_n}$ also span V. (12)

(c) Let $F : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ be a linear map, and let $G : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ be the function defined by

$$G(\overrightarrow{v}) = F(\overrightarrow{v}) + 3\overrightarrow{v}, \quad \forall \overrightarrow{v} \in \mathbb{R}^n.$$

Show that G is also a linear map. (13)

(d) Let $\theta : \mathbb{R}^4 \to \mathbb{R}^3$ be a linear map with

$$\theta((x, y, z, t)) = (2x - z + t, 2y + z, x - y + t).$$

Find a matrix A_θ such that $\theta(\overrightarrow{v}) = A_\theta \overrightarrow{v}$, for every $\overrightarrow{v} \in \mathbb{R}^4$. Furthermore, check your choice of A_θ is correct by showing that $\theta(\overrightarrow{v}) = A_\theta \overrightarrow{v}$ for every vector $\overrightarrow{v} = (x, y, z, t) \in \mathbb{R}^4$. (12)

[50]

SECTION B

2. (a) (i) Recall that there are two standard binary operations on $\mathbb{R}$, $+$ and $\cdot$. State the distributivity property of $\mathbb{R}$.

   (ii) Let z be a complex number and $\{(z_n)\}_{n=1}^{\infty}$ be a sequence of complex numbers. State the definition of $\{(z_n)\}_{n=1}^{\infty}$ *converges* to z. (5)

   (b) Let $x \in \mathbb{R}$ such that $x \neq 0$. Prove that the multiplicative inverse of x is unique. For each step of your proof, state explicitly the axioms that you are applying. (10)

   (c) Let $\sqrt[n]{r}$ denote the n-th root of a positive real number r. For $n = 1$, we define $\sqrt[1]{r} = r$. Prove that (10)

$$\lim_{n\to\infty} \sqrt[n]{n} = 1.$$

[**25**]

3. (a) Let V be a vector space over a field K. Let U be a subset of V. What does it mean for U to be a subspace of V? (3)

   (b) In each of the following cases, determine whether the given subset W is a subspace of V, briefly justifying your answers. (Recall that $\mathrm{Fn}(\mathbb{R})$ is the vector space of real-valued functions on $\mathbb{R}$, and $M_2(\mathbb{R})$ is the vector space of 2×2 matrices over $\mathbb{R}$.) (9)

   (i) $V = \mathbb{R}^3$, $W = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : x^2y^3z^4 = 0 \right\}$;

   (ii) $V = \mathrm{Fn}(\mathbb{R})$, $W = \{f : \mathbb{R} \to \mathbb{R} : f(x) \leq x \text{ for all } x > 0\}$;

   (iii) $V = M_2(\mathbb{R})$, $W = \left\{ \begin{pmatrix} a^2 - 1 & a^3 - 1 \\ b & a + 1 + b \end{pmatrix} : a, b \in \mathbb{R} \right\}$.

   (c) The kernel of a linear map $\theta : V \to W$, denoted $\ker(\theta)$, is defined as

$$\ker(\theta) = \{\overrightarrow{v} \in V : \theta(\overrightarrow{v}) = \overrightarrow{0_W}\}.$$

   (i) Show that $\ker(\theta)$ is a subspace of V. (7)

   (ii) Show that θ is injective if and only if $\ker(\theta) = \{\overrightarrow{0_V}\}$. (6)

[**25**]

4. (a) Let V be a finite-dimensional vector space over $\mathbb{R}$, and let U and W be subspaces of V. Let $\overrightarrow{u_1}, \overrightarrow{u_2}, ..., \overrightarrow{u_r}$ be a basis for U, and let $\overrightarrow{w_1}, \overrightarrow{w_2}, ..., \overrightarrow{w_s}$ be a basis for W. Prove that $U \cap W = \{ \overrightarrow{0} \}$ if and only if $\overrightarrow{u_1}, \overrightarrow{u_2}, ..., \overrightarrow{u_r}, \overrightarrow{w_1}, \overrightarrow{w_2}, ..., \overrightarrow{w_s}$ are linearly independent. (13)

(b) (i) Let $\{a_n\}_{n=1}^{\infty}$ be a sequence of complex numbers and let $k \in \mathbb{N}$. State the definition of the k-th partial sum of this sequence. (2)

(ii) Does the series

$$\sum_{n=1}^{\infty}(-1)^n$$

converge or diverge? Justify your answer. (10)

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